Agentic Social Affordance Framework (ASAF): Agent Identity Design as a Collaboration Interface in Multi-Agent Systems

Meng-Han Lee
Independent Researcher, Taipei, Taiwan
ORCID: 0009-0007-1685-0877
zaious.design@gmail.com

June 29, 2026 | Preprint |

Under review at Frontiers in Computer Science (Human-Media Interaction). This version has been revised in response to reviewer feedback.

1 Abstract

As AI systems evolve from single conversational agents to complex multi-agent architectures, a critical design dimension has been overlooked: how the social identity of individual agents shapes human behavior within the collaboration. This paper introduces the Agentic Social Affordance Framework (ASAF), a theoretical framework that extends Social Affordance theory into the context of multi-agent AI systems. We propose that agent identity design functions not merely as a user interface convention, but as a collaboration interface—structuring how users perceive, approach, and engage with each agent, and thereby influencing the quality of Human-Agent collaboration outcomes. Specifically, ASAF adopts the analytical separability of the social affordance layer and the engineering orchestration layer as a **framing assumption**—an organizing distinction that structures design analysis—rather than as a testable claim about effect-independence. The layers' downstream effects can and do interact; the framing assumption is useful because it identifies a class of design considerations (cognitive posture, role-schema activation, oversight calibration) that is irreducible to engineering choices alone. ASAF comprises three mechanisms: Identity Signaling, Behavioral Priming, and Collaborative Governance, and specifies their boundary conditions through a four-tier Identity Signal Fidelity Spectrum and an individual-difference moderating variable (anthropomorphizing vs. instrumentalizing cognitive style). We situate ASAF in relation to existing affordance theory (Hutchby, 2001), the CASA paradigm (Gambino et al., 2020), and classical multi-agent systems research (Wooldridge & Jennings, 1995), identifying a directional reversal: where classical MAS used roles, norms, and coordination to constrain autonomous agents, ASAF applies the same organizational vocabulary to structure the cognition and oversight of human operators who remain in the loop. ASAF positions social affordance design as a first-class design responsibility that engineering orchestration cannot subsume, with concrete implications for multi-agent system design. We outline directions for future empirical validation, including a factorial design for characterizing the empirical interaction surface between the social affordance and engineering orchestration layers.

Keywords: Social Affordance, Multi-Agent Systems, Human-Agent Interaction, Agent Identity Design, Identity Signal, LLM Agent, Human-in-the-Loop, Cognitive Scaffolding

---

2 1. Introduction

Recent advancements in multi-agent artificial intelligence have driven increased research on orchestrational efficiency, token optimization, and inter-agent communication protocols (e.g., Hong et al., 2023). While this engineering-centric discourse has successfully established the technical scaffolding for collaborative AI, it inherently marginalizes the human-computer interaction (HCI) dimension. Modern multi-agent systems are rarely closed causal loops; they exist primarily in Human-in-the-Loop (HitL) configurations where human operators must direct, parse, and trust the outputs of highly specialized, distinct *node agents*—individual agents occupying defined role-bearing positions within the orchestration topology, each carrying a declared identity and a designated channel of inter-agent communication.

The impetus for this conceptual framework emerged directly from the author's industrial deployment challenges rather than isolated laboratory experiments. During the rapid proliferation of orchestration engines (e.g., LangGraph, AutoGen), the developer community overwhelmingly anthropomorphized agent workflows into corporate team structures (detailed in Section 2.3), focusing intensely on which execution engine to adopt. However, when scaling a custom collaborative system from a single conversational assistant to a scalable multi-agent personal productivity architecture comprising dozens of specialized nodes (the *ChronicleCore* system detailed in Section 4), a critical reality became apparent: the true bottleneck is rarely the underlying orchestration framework. Instead, the fundamental friction point is *Context Governance*: the human operator's capacity to navigate and interact with disparate computational nodes without inducing cognitive collapse or systemic persona drift.

This realization bridges the gap to established literature. Historical meta-analyses in human-robot interaction suggest that human trust and collaborative effectiveness are significantly influenced by an agent's attribute-based design and perceived role (Hancock et al., 2011). In the context of large language models, this dynamic manifests through "role-play," where system prompts cast the model into specific functional and social identities, establishing boundaries for its computational behavior (Shanahan et al., 2023). However, the implications of these role-playing dynamics observed within a cohesive multi-agent topology remain critically under-theorized.

This paper argues that the social identity of individual agents within a network functions not merely as a cosmetic persona, but as a structural collaboration interface. Anthropomorphic architecture in multi-agent systems is neither an emotional user-experience feature nor an arbitrary engineering choice; rather, it is an inevitable design response to human cognitive architecture confronting systems whose scale exceeds unassisted working memory. We adopt working memory capacity estimates (Miller, 1956; Cowan, 2001) as an approximate problem-scale heuristic. While Cowan's measure concerns simultaneous item retention in short-term memory rather than active cross-session navigation, it establishes the cognitive order of magnitude at which unstructured information management becomes untenable; the cognitive offloading literature (Risko & Gilbert, 2016) demonstrates that when task demands approach such capacity limits, users spontaneously seek external scaffolding structures. ASAF posits that at this approximate scale (approximately 4–7 unstructured functional profiles, depending on chunking strategy and individual differences), the application of social schemas transforms from a superficial preference into a structural necessity (i.e., an indispensable design consideration, not a guaranteed effect for all users; effect magnitude remains moderated by individual cognitive style, as specified in H1).

ASAF distinguishes itself from existing theories (such as CASA, classical role theory, and distributed cognition) through an **analytical framing assumption**: the social affordance layer is treated as a class of design considerations that is *not reducible* to engineering orchestration choices. We adopt this framing assumption—rather than claim it as a testable thesis about effect-independence—because the considerations a designer addresses when configuring agent identity (what cognitive posture to signal, what role schemata to activate, what oversight calibration to invite) form a coherent design vocabulary that engineering decisions about orchestration protocols, tool exposure, or inter-agent message passing do not exhaust. The framing assumption is useful because it makes visible a design surface that engineering-centric accounts otherwise leave implicit. The substantive characterization of this design surface is developed below; its empirical consequences—how identity design effects manifest, interact with engineering effects, and are moderated by user cognitive style—are addressed in H1–H3b (Section 5).

To clarify the boundary with CASA (Reeves & Nass, 1996): CASA's core prediction—that humans automatically apply social norms to computers—can account for the basic Identity Signaling effect at the dyadic level (i.e., a single user responding socially to a single personified agent). However, CASA is structurally a dyadic theory; it does not generate predictions about (a) the scaling threshold at which social identity shifts from optional to structurally important (approximately 4–7 agents; H1), (b) the differential oversight calibration across agents occupying distinct topological roles within a system—specifically, that the same agent identity elicits different operator behavior depending on the surrounding agents' role configuration (H3b, Section 5), or (c) the analytical framing of the social design space as a class of considerations not reducible to engineering orchestration choices. CASA-plus-role-schema accounts (Gambino et al., 2020; Biddle, 1986) can generate the basic role-level oversight differential (H3a), but no extension of CASA—including aggregated or conformity-based variants (cf. Song et al., 2025)—generates the topology-level prediction in H3b, because CASA lacks a structural vocabulary for agent-topology-level oversight calibration. ASAF's independent contribution lies precisely in these multi-agent, topology-level, and design-theoretic dimensions that fall outside CASA's explanatory scope.

Under this framing assumption, the social identity design and engineering orchestration design address distinct classes of considerations: configuring agent identity (cognitive posture, interaction norms, role-schema activation) draws on a different design vocabulary than configuring capability (tool exposure, orchestration protocol, message-passing scheme). Engineering restructuring does not entail any particular social identity configuration, and redesigning agent personas does not alter the engineering topology—a separability that is observable at the level of design decisions even when the layers' downstream effects interact. H1 (Section 5) tests one consequence of this framing: whether social identity design carries independent explanatory power beyond functional labeling. The broader interaction surface between the two layers is addressed by the factorial design discussed in Section 7. Consequently, treating multi-agent design purely as a software engineering decomposition problem risks creating structurally coherent but interactionally unreadable systems.

This paper makes the following contributions:

1. We introduce the Agentic Social Affordance Framework (ASAF), distinguishing its predictive power by positioning anthropomorphic identity not as an empathetic illusion, but as a structurally important cognitive load-reduction interface for human operators managing systems that exceed working memory capacity (approximately 4–7 agents, depending on chunking strategy and individual differences). Its effect magnitude is moderated by individual cognitive style (anthropomorphizing vs. instrumentalizing tendency), as specified in H1.
2. We identify and define three core mechanisms through which ASAF operates: Identity Signaling, Behavioral Priming, and Collaborative Governance.
3. We situate ASAF in relation to existing affordance theory (Gibson, 1979; Norman, 1988) and automation reliance literature (Parasuraman & Riley, 1997; Lee & See, 2004).
4. We provide operational design boundaries, formally defining how individual differences in cognitive style (anthropomorphizing vs. instrumentalizing) moderate the framework's predictive validity.
5. We identify an ontological displacement and directional reversal from classical multi-agent systems (Wooldridge & Jennings, 1995): where classical MAS treated human operators—when considered at all—as just another type of agent assimilated into a framework presupposing autonomy, rationality, and stable utility, ASAF applies the same organizational vocabulary to structure the cognition of human operators—theorized as structurally distinct from the agents they oversee—in systems where agents lack the endogenous autonomy that the classical formalism required.

---

3 2. Background and Related Work

3.1 2.1 Affordance Theory: From Gibson to Social Affordance

The concept of affordance was introduced by Gibson (1979) as a relational property between an actor and their environment: the action possibilities that an environment offers to a particular actor. Affordances are neither purely objective properties of objects nor purely subjective constructions of the mind; they emerge from the relationship between the two.

Norman (1988; 1999) operationalized affordance for design contexts, distinguishing between real affordances (what an object actually allows) and perceived affordances (what users believe it allows). Norman's contribution established affordance as a central concept in human-computer interaction, directing attention to how design choices shape user perception and behavior.

Social Affordance extends this framework to the social dimensions of technology. Where Gibson's affordances describe physical action possibilities and Norman's describe functional ones, Social Affordances describe the social interaction possibilities that a system's design signals to its users (Hutchby, 2001). During the Web 2.0 era, this was extensively applied to analyze how the spatial architecture of social network sites shaped "networked publics" and structured human connection (Boyd, 2010). Nagy and Neff (2015) advanced this domain by introducing the concept of *imagined affordances*, arguing that a user's *belief* about what a technology or algorithmic system can do is just as determinant of their behavior as the technology's hard-coded capabilities. This transitioned affordance theory from analyzing static interfaces to analyzing socially constructed expectations.

Recent sociological work has begun bridging this framework into artificial intelligence. A critical conceptual precursor is Sundar's (2020) framework for Human-AI Interaction, which builds on his earlier Theory of Interactive Media Effects (TIME) and examines the tension between human and machine agency. Sundar identifies the "machine heuristic"—where users apply mental shortcuts to evaluate AI outputs based on the mere presence of a machine interface. While Sundar's machine heuristic operates as a binary trigger (machine vs. human), Identity Signaling in multi-agent systems operates as a multi-valued, role-differentiated signal: users do not merely respond to "a machine" but calibrate their behavior to *which kind of social role* the machine presents. This shift from binary agency detection to structured role-based calibration is precisely the granularity gap that ASAF addresses. Empirical studies on ChatGPT usage in specific cultural contexts illustrate how perceived affordances in LLMs are shaped by user interpretation practices rather than fixed machine properties alone (Haqqu et al., 2025). Anthropomorphic design cues further trigger parasocial relationship dynamics, modifying user behavior without changing the underlying model parameters (Reeves & Nass, 1996; Epley et al., 2007). Together, these findings indicate that affordance theory must evolve past simple "machine heuristics" toward interpreting the explicit role schemata embedded in system design.

3.2 2.2 The Chatbot Era: Single-Agent Social Affordance

In the Chatbot era, Social Affordance operated through a single, often ambiguous social identity. ChatGPT, Claude, and similar systems present a unified interface that affords many things—assistance, conversation, analysis, creativity—but commits to none. This ambiguity is a design choice that maximizes perceived utility but minimizes Social Affordance specificity. Users engage with these systems through general-purpose interaction patterns, rarely adjusting their behavior based on a perceived social role of the system.

The limitation is structural: with a single agent identity, Social Affordance can only operate at the level of the whole system. There is no mechanism for users to adjust their interaction style based on the specific capability or social role of a particular component.

3.3 2.3 The Evolution of Agentic Roles: The "Virtual Team" Wave and Its Backlash

Multi-agent AI systems introduce a structurally distinct architecture, replacing single conversational models with networks of specialized agents. Beginning in late 2025 and accelerating through the first quarter of 2026, a rapid resurgence of anthropomorphic multi-agent architectures emerged, what we term here the **Virtual Team Wave**. Developers returned to structuring agents as corporate role hierarchies, and in some cases, historical administrative metaphors. Three representative open-source implementations collectively accumulated **169,732 GitHub stars** within this window, surfacing an apparent industry-wide appetite for persona-structured agent fleets.[1]

These implementations illustrate both the scale of the trend and the cross-cultural nature of the underlying design impulse. *The Agency* (Sitarzewski, 2025), which seeded the wave in October 2025, organizes 147 agents into corporate-style divisions (engineering, design, marketing, product), each agent written as "a specialized expert with personality, processes, and proven deliverables." *gstack* (Tan, 2026), authored by the President of Y Combinator and released in March 2026, operationalizes a single builder's team as "23 opinionated tools that serve as CEO, Designer, Eng Manager, Release Manager, Doc Engineer, and QA," accumulating over 73,000 stars within its first five weeks. *Edict* (cft0808, 2026; the maintainer publishes under this GitHub handle) departs from the Silicon Valley corporate metaphor entirely, adopting the **Three Departments and Six Ministries** (三省六部) system of Tang Dynasty governance (ca. 618 CE). It distributes agent roles across the Secretariat, the Chancellery, and the Department of State Affairs, operating under a strict inter-agent permission matrix. That the same underlying design impulse surfaces in both Silicon Valley startup vernacular and 1,400-year-old administrative structure suggests that this anthropomorphization is not a local stylistic preference but a recurring cognitive response to the orchestration demands of multi-agent systems. At the commercial frontier, xAI's Grok 4.20 (released February 2026) deploys four named specialist agents as its default inference architecture (xAI, 2026).[2] This structural shift towards roleplaying clusters is not an isolated design anomaly but an emergent standard practice. Qualitative studies of early adopters confirm that practitioners organically conceptualize and interact with complex LLM pipelines

[1] Repository metrics retrieved via the GitHub REST API (GET /repos/{owner}/{repo}) on 2026-04-16. Aggregate at retrieval: 169,732 stars, 24,954 forks across the three repositories. These figures aggregate three distinct repositories; actual unique observer count is lower due to overlap among early-adopter communities who star multiple related projects—a sampling bias inherent to GitHub stars as a social signal inflated by high-profile distribution events (e.g., curator endorsements). The metric is cited as a directional indicator of industry interest, not as a precise adoption measure. Independent of open-source enthusiasm, enterprise deployments in Q1–Q2 2026 have begun embedding agent identity into infrastructure layers (e.g., Microsoft Entra Agent ID, Google Agent Registry, AWS Bedrock AgentCore), suggesting that role-structured agent architectures are transitioning from experimental practice to production infrastructure.

[2] xAI's Grok 4.20 (released February 17, 2026) markets "four expert-mode AI agents" as a core feature of its paid SuperGrok tier (grok.com). The individual agent names—including Benjamin (mathematical reasoning) and Lucas (contrarian analysis)—are observable within the paid product interface (SuperGrok subscription required).

as "teams" of specialized, task-based collaborators rather than mere software threads (Naik et al., 2025).

However, while Naik et al. (2025) establish *that* early adopters default to this team-oriented mental model, the underlying mechanism shaping interaction remains untheorized. Empirical work concurrently demonstrates that groups of AI agents can function as cohesive social groups, exerting normative social pressure on users through mechanisms analogous to human group dynamics (Song et al., 2025).

Recent empirical work has demonstrated that specific agent properties (e.g., group presence (Song et al., 2025), verbalized uncertainty (Xu et al., 2025), and confidence calibration (Li et al., 2024)) exert measurable influence on human collaboration behavior. However, these findings have yet to be integrated under a unifying design-level framework. ASAF offers such an integration by treating each of these properties as instances of Social Affordance signaling.

3.4 2.3.1 Situating ASAF Against Classical Multi-Agent Systems and CASA

The organizational vocabulary that ASAF deploys—roles, norms, coordination, governance—is not new; it descends directly from classical multi-agent systems (MAS) research (Wooldridge & Jennings, 1995; Smith, 1980) and the normative MAS tradition (Dignum, 2004). In classical MAS, agents were assumed to possess endogenous autonomy—entities capable of independently forming intentions, commitments, and action plans (Rao & Georgeff, 1995). Roles, norms, coordination protocols, and filtering mechanisms were therefore designed to constrain these autonomous agents into coherent collective behavior.

Modern LLM-based agents invert this assumption. They exhibit strong generative capability while their coherence as persistent role-occupying agents is externally supplied through orchestration, persistent memory, and capability boundaries. This externalization is not merely a function of current model limitations—it is also a deliberate architectural choice for reasons of governability, auditability, and predictability, and remains a viable design pattern independent of underlying model capability. ASAF's predictive scope is bounded to architectures where this externalization is the operative condition (Tiers 1-3 of the Fidelity Spectrum, Section 2.3; deployment tier reflects operational requirements rather than technological maturity). When classical MAS considered human operators at all, they were assimilated as one type of agent within the same formal framework—endowed with the same autonomy assumptions, communication protocols, and rationality requirements as artificial agents. The emergence of Human-Agent Interaction as a distinct research tradition—rather than subsuming human interaction within MAS's agent formalism—itself evidences this assimilation gap; qualitative studies of early adopters further confirm that practitioners organically conceptualize LLM pipelines as teams of specialized collaborators rather than mere software threads (Naik et al., 2025), a conceptualization that classical MAS's formalism was not designed to accommodate.

Emerging critiques of LLM-based MAS have begun to identify this ontological displacement from empirical, information-theoretic, and cognitive grounds. ASAF does not rehearse these critiques. Rather, it identifies their shared consequence at the cognitive layer: when the agent ontology presupposed by classical MAS fails to transfer, the human operator—systematically excluded from classical MAS theory or assimilated as just another agent—becomes the system's actual coordination bottleneck. ASAF therefore retains the MAS organizational vocabulary—roles, norms, coordination, governance—but inverts its directional target. This reversal—from agent-facing constraint to human-facing cognitive interface—is what distinguishes ASAF from classical MAS re-description.[3]

ASAF must also be distinguished from the Computers Are Social Actors (CASA) paradigm (Reeves & Nass, 1996) and its multi-source extensions (Gambino, Fox, & Ratan, 2020). CASA and its extensions treat social responses to machines as automatic and mindless—a cognitive default that occurs regardless of design intent. Gambino et al. (2020) extend this to media agents, arguing that users develop specific scripts for human-media agent interaction that are, like human-human scripts, applied mindlessly. ASAF operates in a condition that CASA did not anticipate: LLM-based agents whose conversational performance is sufficiently capable that role-based interaction is not a cognitive bias to be explained away, but a functionally productive design strategy to be deliberately engineered. In this condition, the question shifts from "why do users treat machines as social actors?" (CASA) to "how should designers structure agent identity to optimize human-agent collaboration?" (ASAF). Furthermore, ASAF's predictions operate at the system topology level—the same agent identity elicits different operator behavior depending on the topological role configuration of surrounding agents (H3b, Section 5)—a structural property that dyadic frameworks, including multi-source CASA extensions, do not generate.

ASAF must also be distinguished from the human-agent teaming tradition (mixed-initiative interaction, coactive design, and adjustable autonomy). These theories address how a human and one or more agents partition initiative and dynamically adjust autonomy during task execution; their basic unit is the human–agent relationship, whether one agent or several under a single operator's supervision. ASAF addresses a structurally different object: how agent identity design structures the operator's cognitive load and differential oversight across a multi-agent fleet. Crucially, the teaming unit—however many agents it scales to—does not represent how the agents' identities are configured relative to one another, because its formalism models the human–agent link rather than the inter-agent identity topology. The two frameworks are therefore not mutually exclusive; they describe distinct layers of human–agent interaction (autonomy partition within human–agent links vs. identity-driven oversight across the agent topology) and may co-apply within the same system. The empirical signature isolating ASAF's layer is H3b (Section 5): the same agent identity elicits different operator oversight as a function of the surrounding agents' identity configuration—a topology-level prediction that frameworks built on the human–agent link, including the teaming tradition, do not structurally generate.

However, within the same window a fierce **Anti-Roleplay Backlash** emerged in parallel.

[3] La Malfa et al. (2025), including Wooldridge, diagnose a similar ontological mismatch but prescribe closer alignment with classical MAS principles—a prescription ASAF does not share, as it treats the ontological gap not as a deviation to be corrected but as a structural feature of LLM-based agents that demands a new theoretical framework rather than stricter adherence to the old one.

The developer community identified a fatal architectural flaw in the "virtual team" model: *information decay during transmission*. Unlike humans who compensate for communication gaps with shared cultural context, LLMs passing synthesized documents down a rigid assembly line suffer from continuous context loss and "reasoning drift." Artificially imposing strict role boundaries restricts the model's inherent cross-domain capabilities, creating false walls precisely where valuable edge-case reasoning tends to occur.

As a result, engineering consensus has swung violently away from persona-driven pipelines. Leading infrastructure providers now explicitly warn against fragmented agent architectures (e.g., Cognition AI's critique of multi-agent fragility; Yan, 2025), prioritizing pure deterministic paradigms instead. Frontier labs such as Anthropic (2025a), for instance, have publicly advocated "Context Engineering" and the use of explicit shared state files (e.g., persistent `progress.txt` or Git-based logs) over sequential roleplay. In this current paradigm, agent identity is widely dismissed as a dangerous, anthropomorphic illusion.

ASAF's mechanisms presuppose that agent identity signals remain perceivable to the human operator across interactions. To contextualize ASAF's scope against engineering critiques of context decay (e.g., Yan, 2025), we structure agent identity persistence along a four-tier design spectrum:

1. **Pure Prompt Injection**: Personas are defined per-session with no persistence, rendering them highly susceptible to reasoning drift.
2. **Prompt with Persistent Memory**: Personas span sessions via memory access, but lack structured cross-agent auditing.
3. **Structured Identity Enforcement**: Agents possess persistent structural modules, hard-gated orchestration layers, and cross-agent validation mechanisms (e.g., the ChronicleCore architecture detailed in Section 4).
4. **Model-Driven Routing**: Specialist routing occurs natively at inference-time and cannot be bypassed by the user (e.g., Grok 4.20).

ASAF scopes its strongest predictions to systems operating at Tier 3 and 4, where Social Affordances are structurally enforced. The framework's predictions apply partially to Tier 2, while Tier 1 serves as a boundary condition where framework effectiveness degrades proportionally with identity signal fidelity. Yan's (2025) critique accurately describes the structural fragility inherent to Tier 1 and 2 deployments. However, his critique targets the engineering layer (context fragility), while ASAF operates on the cognitive layer (how human operators navigate deployed architecture). These are analytically separable concerns: configuring engineering robustness and configuring agent identity draw on distinct design vocabularies, even when their downstream effects interact (formalized as the analytical framing assumption in Section 3.1). The appropriate tier for any deployment is determined by operational requirements and risk tolerance, not by technological epoch. ASAF's predictive validity scales with identity signal fidelity across this spectrum—its mechanisms remain conceptually applicable at any tier, with strongest empirical predictions at Tier 3-4 where structural enforcement removes identity drift as a confound.

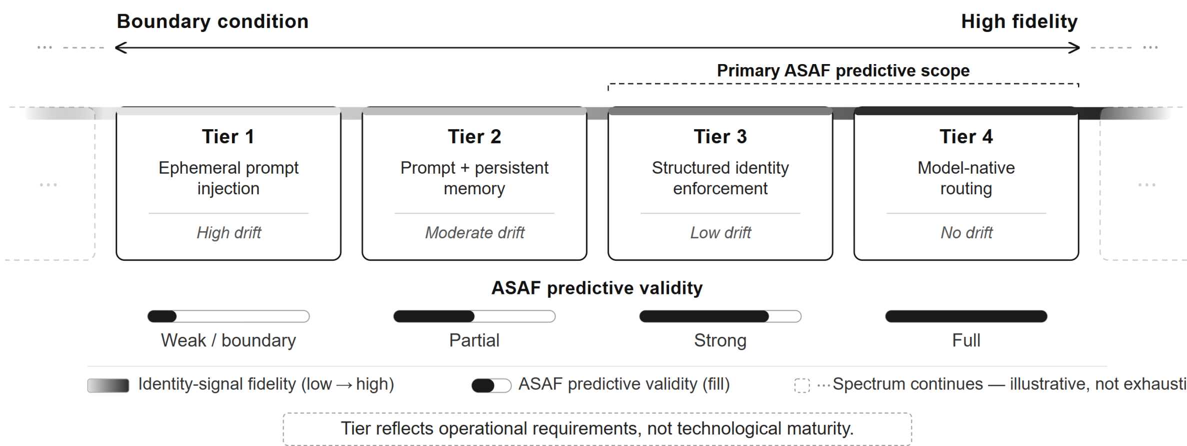

Figure 1: The Identity Signal Fidelity Spectrum. ASAF's predictive validity scales with the structural enforcement of agent identity signals. Tier 1 (ephemeral prompt injection) represents a boundary condition where identity drift degrades framework predictions proportionally. Tiers 3–4, where identity signals are structurally enforced or model-native, constitute the primary scope of ASAF's strongest empirical predictions. The four tiers are illustrative rather than exhaustive—sampled points along a continuous spectrum (indicated by the fading band and open ends), not a closed enumeration. The spectrum is not a developmental trajectory—deployment tier is determined by operational requirements, not technological maturity.

3.5 2.4 The Cognitive Interface Imperative

Building upon early theories of "cognitive technologies" that reorganize human mental functioning (Pea, 1985; Salomon, 1993), and on the growing literature on cognitive offloading in everyday tool use (Risko & Gilbert, 2016), we conceptually define a well-designed multi-agent fleet as an external cognitive scaffold. A system whose structure mirrors how the human architect's brain naturally decomposes problems externalizes operations that would otherwise exhaust working memory. When each agent carries a legible social identity, this scaffolding effect becomes structured and predictable rather than accidental (Amershi et al., 2019). The potential for deliberate Human-Agent cognitive synergy of this kind remains largely absent from current orchestration literature.

Reducing multi-agent architectures to purely functional software pipelines ignores the true operational bottleneck of Human-in-the-Loop systems. The ultimate hardware ceiling of any highly scaled agentic system is not GPU capacity or API rate limits; it is the *human operator's own cognitive architecture*.

Foundational HCI research is unambiguous here: effective system design must align with human mental models, not just maximize output accuracy (Bansal et al., 2019). In the context of LLMs, Shanahan et al. (2023) frame these models as "simulators of text-generating processes" capable of instantiating diverse "simulacra" (personas or roles). Agent Identity, therefore, acts as an **Interpretability Prior**: by casting the underlying simulator into a distinct social role, a developer constrains behavioral unpredictability while simultaneously giving the human operator a familiar heuristic for predicting the agent's "error boundaries" and epistemic scope.

A critical boundary condition of ASAF is its explicit dependence on individual differences in cognitive style as a formal moderating variable. Specifically, the tendency to *anthropomorphize* versus *instrumentalize* computational tools (Epley et al., 2007) moderates the magnitude of the framework's predicted effects. ASAF remains theoretically applicable across the entire user spectrum, but the effect size of identity design on cognitive offloading is directly proportional to a user's natural propensity to employ role schemata. For users who adopt a strictly instrumentalist stance (actively resisting role-schema activation and treating agents purely as mechanistic software threads), the cognitive scaffolding effect is significantly attenuated. For these instrumentalizing users, encountering role-based design at scale (approximately 4–7 agents) may introduce cognitive overhead rather than reduce it, as they force themselves to track raw functional boundaries rather than relying on intuitive social heuristics. By positioning anthropomorphic tendency as a moderating variable rather than an absolute structural prerequisite, ASAF generates distinct, falsifiable interaction trajectories for mixed user populations.

---

4 3. The Agentic Social Affordance Framework (ASAF)

ASAF proposes that in multi-agent AI systems with Human-in-the-Loop configurations, agent identity design functions as a Social Affordance layer. Before detailing its operational mechanisms, it is critical to anchor this framework in its sociological and cognitive psychology origins.

4.1 3.1 Theoretical Foundations: Cognition, Sociology, and Co-construction

In the context of ASAF, a *Social Affordance* is operationally defined as a relational property of an agent's designed identity that signals to a human user the interaction possibilities, epistemic scope, and appropriate collaboration modality specific to that agent's declared role. Following Hutchby's (2001) "third way" between constructivist and realist accounts of technology, ASAF treats agent identity as neither a purely designer-imposed property (realism) nor an arbitrary user projection (constructivism), but a relational affordance that emerges between designed identity signals and users' pre-existing role schemata. The relational character of social affordance entails that the same identity label may afford different interaction possibilities depending on the organizational, professional, and cultural context in which it is encountered (Hutchby, 2001). ASAF's predictions are therefore strongest for identity labels whose core cognitive posture is cross-contextually stable—roles such as auditor, critic, or peer reviewer carry structurally similar behavioral expectations across cultural and institutional contexts, because their core cognitive posture (challenging, demanding evidence, withholding acceptance) is defined by professional function rather than local cultural convention. Consequently, ASAF's claim that the social affordance layer constitutes an independent design dimension (Section 1) is an *analytical framing assumption*: the social layer and the engineering layer represent distinct classes of design considerations, drawing on distinct design vocabularies. The framing assumption is positive rather than definitional—it identifies a coherent set of design questions (what cognitive posture to signal, what role schemata to activate, what oversight to invite) that engineering decisions about orchestration, tool exposure, and message passing do not exhaust. It is not a thesis about effect-independence: the realized magnitude of any Social Affordance is inherently user-moderated, scaling with the individual's propensity to activate role schemata (see H1, Section 5), and the layers' downstream effects can and do interact (see Section 3.4 on persona-capability interaction). The distinction is substantive—it concerns the structure of the design problem—but it organizes design analysis rather than issuing a prediction about effect-independence.

A clarification on the relationship between agent identity and social scripts is necessary. On the supply side, agents *enact* culturally established role sequences in the sense characterized by Schank and Abelson (1977)—stereotyped action patterns with defined roles, scenes, and behavioral expectations—drawn from professional stereotypes, narrative media, and cultural archetypes. We use "script" here in a descriptive rather than cognitive sense: the agent does not cognitively possess a script, but its designed behavior instantiates a culturally recognizable action sequence. On the demand side, however, users do not necessarily require pre-existing scripts for interacting with these enacted roles. What enables interaction initiation is social affordance perception (Hutchby, 2001): the user recognizes the agent as the locus where a particular cognitive function—interrogation, verification, ideation—can be invoked. Identity Signaling (Section 3.2) serves as the bridge between these two sides, translating the agent's enacted role into a perceivable affordance for the human operator. The initial affordance provides a cognitive entry point; the specific interaction pattern is subsequently co-constructed through sustained use.

ASAF's treatment of agent identity as an "Interpretability Prior" inherits the instrumentalist stance on intentional description articulated by McCarthy (1979) and operationalized for agent theory by Wooldridge and Jennings (1995): attributing roles and social identities to agents is legitimate not because agents possess these properties, but because such attribution reduces the cognitive burden on human operators managing systems whose internal structure is insufficiently transparent for mechanistic description. ASAF extends this from a descriptive tool for understanding individual agents to an operational interface for collaborating with agent collectives. Therefore, the efficacy of ASAF does not depend on the LLM possessing actual sociological awareness, but on the human operator possessing culturally available role schemata that agent identity design can activate. We anchor this claim in three theoretical pillars:

First, sociological **Role Theory** (Biddle, 1986) establishes that societal roles carry specific "role expectations." ASAF does not assume that role expectations activated by agent identity labels are universal or automatically shared. Following Biddle's integrative account, ASAF treats initial role expectations as structurally primed—entering an interaction with an agent labeled "Auditor" activates a default behavioral script comparable to entering a hospital and encountering a "doctor"—but subject to ongoing negotiation through use. Users routinely recalibrate agent identities through continued interaction, from adjusting prompt specificity to modifying persona configurations, a process enabled by the low technical threshold of LLM-based systems. The initial role schema provides a cognitive entry point; the enacted role is co-constructed through sustained operator-agent interaction.

Second, Goffman's **Dramaturgical Analysis** (1959) introduces a structural distinction between front stage (the performance presented to audiences) and back stage (the private space where performances are prepared). In LLM-based agent systems, this distinction acquires a distinctive character: the back stage—the computational process generating responses—is opaque to the human operator regardless of design choices. Whether or not this process constitutes "understanding" is an open question (Shanahan et al., 2023); what is not open is that the human operator cannot access it. The front stage—the agent's identity as presented through name, role, tone, and behavioral constraints—is therefore not a mask concealing a "true self" behind it, but the operationally primary interface through which the human operator constructs expectations, calibrates trust, and negotiates the interaction. As in Goffman's original framework, this front stage is not unilaterally imposed by the designer; it is co-constructed through sustained interaction, with users continuously recalibrating their interpretation of the agent's role.

Finally, **Distributed Cognition** (Hutchins, 1995; 2010) models how cognitive labor is redistributed across systems comprising human, artifactual, and representational components. Critically, Hutchins demonstrates that distribution does not eliminate cognitive labor but redistributes it—and the coordination required to maintain distributed processes is itself a form of cognitive work (Hutchins, 2010). ASAF draws on this insight with a specific qualification: identity-based role schemata do not provide automatic cognitive routing, but they reduce the coordination cost of navigating a multi-agent system by providing familiar heuristic entry points for each node. The coordination labor remains—users must still calibrate trust, verify outputs, and negotiate role boundaries—but identity signals lower the threshold for initiating that coordination, particularly when the number of agents exceeds unaided working memory capacity.

These three foundational theories manifest in ASAF through three distinct but conceptually interdependent mechanisms, operating across three levels of analysis. Identity Signaling functions at the cognitive level, shaping how an individual user approaches a single agent before any interaction occurs. Behavioral Priming functions at the interactional level, describing how that initial framing propagates into sustained changes in user input behavior across the session. Collaborative Governance functions at the system topology level, operating as a bridging concept across two temporal dimensions. At *design-time*, establishing deliberate identity structures (e.g., configuring adversarial roles) serves as the architectural prerequisite that enables distinct Social Affordances to exist. At *run-time*, this topology governs the emergent dynamics—specifically, how human operators differentially calibrate their oversight based on the cascaded effects of Identity Signaling and Behavioral Priming. This structural interplay establishes the human cognitive interface in multi-agent systems as an independent design dimension, standing alongside the engineering orchestration layer rather than subordinate to it.

4.2 3.2 Mechanism 1: Identity Signaling

Definition: Agent identity design activates role schemata in users before interaction begins, reducing cognitive friction and establishing structural interaction expectations.

Recent scholarship in HCI highlights the growing necessity of "thoughtful persona design" in LLMs, predominantly focusing on elements like voice, empathy, and emotional demographics to enhance single-agent interaction quality (Zargham et al., 2024). ASAF diverges from this paradigm. In single-agent contexts, Social Affordance operates primarily through emotional rapport and conversational empathy—a diffuse, system-level property with no comparative anchor. In multi-agent contexts, this function undergoes a scale-dependent reprioritization: as users must navigate multiple distinct agents, structural boundary-marking between differentiated identities moves to the foreground, while emotional rapport recedes—not stripped, but reprioritized. This reprioritization is consistent with Hutchby's (2001) relational account: the same technological artifact affords different interaction possibilities depending on the complexity and structure of the context in which it is encountered.

An Identity Signal, as used in this framework, is characterized by two operationally distinct properties. *Signal intensity* refers to the degree to which the identity design specifies a precise cognitive posture—the difference between "you are an assistant" and a fully articulated professional archetype with defined judgment criteria, rhetorical style, and epistemic boundaries. *Signal fidelity* refers to the degree to which the agent's enacted behavior remains consistent

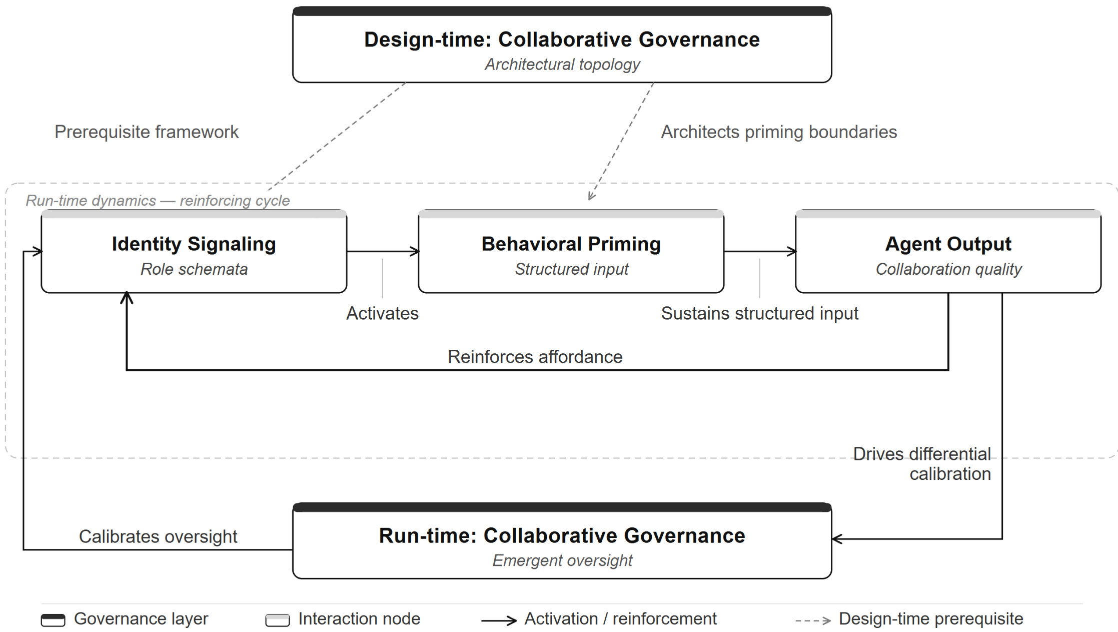

Figure 2: The dual dimensions of Collaborative Governance. Design-time topology acts as the architectural prerequisite for interaction, while run-time emergent governance reflects the human operator's differential oversight. Within the run-time cycle, Identity Signaling activates role schemata and Behavioral Priming sustains structured input; Agent Output reinforces the active affordance, and drives differential calibration of oversight back into the governance layer.

with the declared identity across interactions, maintained through persistent memory mechanisms and structural enforcement. These two properties vary independently: a system may deploy high-intensity signals with low fidelity (vivid personas that drift within a session) or high-fidelity signals with low intensity (stable but vaguely defined roles). On the supply side, signal intensity is a function of how precisely the agent's enacted social script (Section 3.1; Schank & Abelson, 1977) specifies a recognizable cognitive posture; on the demand side, it is a function of how readily the user's role schemata are activated by that specification. The Identity Signal Fidelity Spectrum (Section 2.3, Tiers 1–4) classifies the technical architecture by which fidelity is maintained—the minimum structural enforcement required for identity signals to operate reliably—rather than properties of the signals themselves.

ASAF posits that when a user encounters an agent with a defined identity (for instance, an architecture deploying "Benjamin, specializing in mathematical deduction" alongside "Lucas, a dedicated contrarian analyst"), they do not begin interaction from a blank slate. We theorize that they bring to bear culturally available role schemata for interacting with entities who occupy similar analogous roles. These schemata implicitly afford a sense of what kind of input is appropriate, what level of formality is expected, and what type of output to anticipate.

This mechanism predicts that social identity will outperform functionally equivalent but non-personified labels (H1: condition a $>$ b) for three specific reasons. First, *activation automaticity*: role schemata are acquired through a lifetime of interpersonal interaction, media consumption, and professional socialization, and activate involuntarily upon encounter (Epley et al., 2007), whereas functional descriptions (e.g., "logical analysis module") require deliberate parsing to derive appropriate interaction norms. Second, *default coverage breadth*: a social role implicitly encodes not only capability boundaries but also tone expectations, formality level, error tolerance, and appropriate challenge behavior—behavioral norms that a functional label does not carry. Third, *normative load*: social roles carry implicit behavioral obligations for *both parties* in the interaction (Biddle, 1986); a user interacting with a "critic" feels normatively compelled to present defensible arguments, an effect that a label reading "verification subsystem" does not generate. These three properties collectively predict that social identity signals will produce richer behavioral adaptation than equivalent functional information alone.

This pre-interaction activation is the core of Identity Signaling. The Social Affordance operates at the moment of encounter, structuring the user's approach to the interaction and strategically reducing the cognitive effort required to determine "how should I context-prompt this system?" This creates a specific scaling advantage in multi-agent environments. When managing approximately 4–7 agents—the heuristic range at which unstructured functional profiles approach working memory limits (Cowan, 2001)—agent identities sidestep these limits by engaging pre-existing social schemas (Epley et al., 2007) rather than requiring the operator to manually track abstract architectural boundaries.

This pre-interaction activation also resolves ambiguities in how users perceive agent output parameters. Consider the counterintuitive finding by Xu et al. (2025), who demonstrated that expressing medium uncertainty (e.g., "I think this might be…") frequently outperforms absolute confidence in human-AI teaming. Seen through the lens of Identity Signaling, this makes complete sense. We do not cognitively expect an "ideator" to offer design propositions with clinical precision or formal rigor. When such an identity signals uncertainty, the user interprets it not as a system failure, but as a culturally legible invitation to step in and exercise human judgment.

4.3 3.3 Mechanism 2: Behavioral Priming

Definition: Agent social identity is anticipated to systematically improve the structural quality of user inputs, creating a positive feedback loop that elevates collaboration outcomes.

Jahani et al. (2026) document how users iteratively learn to adapt their prompting behaviors over time, with significant performance gains emerging from this acquired interaction fluency rather than from model improvements alone.

ASAF's Behavioral Priming mechanism theorizes how agent identity drives this influence. We postulate that when interacting with what they tacitly treat as a domain-competent counterpart (cf. Reeves & Nass, 1996), users are primed to voluntarily: * Provide more structured, context-rich inputs aligned directly with the agent's declared domain. * Proactively include constraints, assumptions, and boundary conditions. * Formulate questions at a higher level of structural specificity.

Jahani et al. (2026) document that this fluency typically develops through iterative trial-and-error across multiple sessions—a process they characterize as a *dynamic complement*, noting that adaptation imposes real costs on users and that failure to adapt degrades outcomes. ASAF hypothesizes that strong Identity Signaling may reduce the initial adaptation cost by providing role-appropriate cognitive entry points from the first interaction, though the extent of this compression remains an empirical question.

A secondary, yet crucial implication of Behavioral Priming relates to cognitive bias mitigation. It is well-documented that human users are dangerously susceptible to AI overconfidence, often failing to detect when a model is hallucinating assertively (Li et al., 2024). However, what happens if we deliberately assign the system a socially cautious or adversarial persona? If a user explicitly knows they are interacting with an "auditor" (by definition, a role whose enacted behavior foregrounds challenge and verification over generative output), this alone structurally intercepts their default vulnerability to over-reliance. The social frame does the work; no explicit instruction to "be skeptical" is required. We further note that in LLM-based systems where designer and operator are coupled, Behavioral Priming co-evolves with operator learning—a structural condition we treat as a feature of LLM agent research methodology rather than a confound to be controlled away (elaborated in future work).

4.4 3.4 Mechanism 3: Collaborative Governance

Definition: In Human-in-the-Loop configurations, agent social identity structures the location and mode of human intervention, transforming HitL from a mere safety checkpoint into a collaboration design element.

The conventional framing of Human-in-the-Loop in AI systems is safety-centric: human oversight exists purely to catch errors. This framing is correct but incomplete. The foundational literature on automation reliance (Parasuraman & Riley, 1997) and the dynamics of human trust in automated systems (Lee & See, 2004) establishes that human operators constantly modulate their reliance and oversight based on the system's perceived purpose and past performance. ASAF extends this to the multi-agent context, positing that when a system possesses defined agent identities, users do not intervene generically; their oversight calibration is dynamically structured by each agent's specific social role. This governance operates across two temporal scopes. At *design-time*, the framework prescribes verifying that newly introduced agents activate distinct role schemata rather than providing redundant Social Affordance signals. At *run-time*, it prescribes monitoring for epistemic convergence between agents whose identities were intended to remain distinct. Section 4.2 operationalizes both functions in the ChronicleCore implementation. In a recent scoping review of 134 top-tier HCI papers, Zhang et al. (2025) constructed an integrated theoretical map of how "agency" and control are distributed in human-AI co-creation. ASAF's Collaborative Governance mechanism serves to operationalize the specific control dynamics mapped by Zhang. We argue that assigning a social identity to an agent is a specific method of *agency negotiation*. A user understanding that one agent is an "ideator" and another is an "auditor" will trust the ideator's divergence while heavily scrutinizing the auditor's final checks, thereby calibrating how deeply to intervene at each stage of the workflow.

This mechanism extends into group dynamics. Empirical research validates that groups of AI agents can function as social groups, exerting "normative social pressure" on human users, particularly through conformity effects when multiple agents agree (Song et al., 2025). Under ASAF's Collaborative Governance, this pressure can be architecturally countered. By deliberately designing interacting agents with competing identities (e.g., assigning a "Devil's Advocate" identity to explicitly challenge a "Proposer" identity), developers can artificially break the unanimity that causes dangerous conformity effects, structurally forcing the human operator to exercise critical judgment. The governance structure of the system becomes legible and manipulable through the Social Affordance layer. What was once invisible coordination logic is now a perceivable, interactable interface that the human operator can navigate directly.

A critical in-scope boundary condition of Collaborative Governance is the **false-specialization inversion**. When Identity Signaling produces false confidence in agent specialization that the underlying model cannot deliver—for example, an agent titled "Data Verification Analyst" that hallucinates frequently—the social affordance *inverts*. Under ASAF's own governance predictions, this inversion produces worse trust miscalibration than if no identity had been assigned at all, because the user's oversight has been calibrated to a social contract the agent cannot honor. This false-specialization inversion addresses Yan's (2025) critique of multi-agent fragility from within ASAF's framework: Yan's concern that artificial role boundaries induce reasoning drift is partly cognitive—it concerns the misalignment between operator mental models, primed by role assignment, and the model's actual capability surface. ASAF therefore predicts both when identity design helps and when it actively harms. In LLM-based systems, persona specification and behavioral capability exhibit interaction effects—persona instructions can shift the distribution from which agent responses are sampled. The framework treats analytical separability as an organizing assumption about *design vocabularies*; the design vocabularies remain distinct (configuring "what cognitive posture to signal" is a different question from configuring "what tools to expose") even though the resulting system properties—including how persona instructions reshape the sampling distribution—are jointly determined.

Taken together, the preceding mechanisms make explicit that ASAF predicts three categories of anthropomorphism-related risk that are integral to the framework rather than peripheral concerns. First, *cognitive overhead risk*: for users with strong instrumentalizing tendencies, role-based design at scale (approximately 4–7 agents) imposes interpretive cost rather than reducing it (Section 2.4). Second, *trust miscalibration risk*: the false-specialization inversion formalized above predicts that misleading identity signals produce worse oversight outcomes than the absence of identity assignment, because operator oversight has been calibrated to a social contract the agent cannot honor. Third, *overconfidence-attenuation failure*: ASAF positions adversarial or auditor-typed identity signals as a structural intervention against user susceptibility to AI overconfidence (Section 3.3), with the corollary that omitting such identity scaffolding leaves operators exposed to default overreliance patterns documented in the human-AI trust literature (Lee & See, 2004; Li et al., 2024). These three categories are core ASAF predictions, not externalities to be deferred. The boundary phenomena discussed in Section 7 (parasocial dynamics, cross-cultural variation, dynamic identity formation) represent unmeasured outer surfaces of these core risks rather than substitutes for them, and constitute the priority directions for empirical extension.

---

5 4. System Implementation: The Design Problem That Motivated ASAF

Rather than presenting empirical validation, we offer the development of *ChronicleCore*—an evolving, single-operator Multi-Agent System—as **the design problem that motivated ASAF**: a case that illustrates how the framework's design space might be navigated in practice, and that surfaces the questions ASAF aims to theorize. Validation of the framework's predictions is explicitly deferred to the empirical designs specified in H1–H3 (Section 5). The researcher simultaneously occupied the roles of system designer, primary operator, and observing analyst. This configuration is useful for generating design rationale and surfacing tacit operational knowledge, but it does not constitute independent empirical validation—a limitation the author acknowledges without reservation.

This paper was itself produced under a multi-agent AI collaboration configuration (see Acknowledgments for specific tools and versions). All conceptual contributions (the ASAF framework, its mechanisms, and hypotheses) originated entirely from the human author and all cited works were manually verified prior to inclusion. The author acknowledges the circular evidentiary structure this creates—using a multi-agent configuration to argue for the value of multi-agent configurations. This methodological reflexivity is part of the framework's boundary conditions rather than its validation.

At the time of writing (April 2026), ChronicleCore comprises 38 active Human-in-the-Loop expert nodes, designed to organically accommodate incremental agent addition as operational needs dictate. Operated by the author as a personal productivity architecture, this case illustrates the design challenges that arise when a single operator must navigate a system whose scale exceeds unaided working memory. Unlike multi-agent simulations of human societal behavior (Park et al., 2023), ChronicleCore utilizes deliberately designed societal structures (agent identities) to structure the reasoning pathways of a single human operator.

5.1 4.1 Decoupling Agents by Epistemic Role

When a single operator scales a system to several dozen specialized agents, the primary challenge transcends tool orchestration and enters the realm of personal cognitive governance. A single, monolithic chatbot affords generic conversational prompting, but a highly populated agent network risks cognitive overload for the human operator and "persona drift" among the agents themselves. To counteract this, ChronicleCore physically decouples the system into distinct operational pillars, each functioning as a highly specific Social Affordance boundary.

Rather than deploying agents as indistinguishable utility nodes, the architecture categorizes them by their epistemic and rhetorical constraints. The system separates strategic routing nodes (which allocate global context but possess no execution capabilities) from sensory nodes responsible purely for data ingestion. It isolates aesthetic alignment tasks from adversarial verification layers. By granting each agent localized context paired with a heavily defined social identity (e.g., naming a verification node the "Inquisitor"—see Appendix A for a summary of the agent identity definition and Appendix B, §B.3 for the full Iron Laws table), the system is designed to structurally force the human operator to calibrate their oversight. This architectural decoupling was designed precisely to shape the operator's interaction patterns. By structurally enforcing the separation—where an adversarial node is configured (through prompt instruction and capability gating) to focus exclusively on probing logical vulnerabilities rather than generative code production—the system aims to preempt persona drift (subject to the persona-capability interaction effects acknowledged in Section 3.4). The design enforces the boundary, ensuring that the operator cannot bypass the intended Social Affordance without breaking the system's operational protocol.

5.2 4.2 Sustaining Identity Boundaries: A Design Perspective

The persistence challenge in a multi-agent system with differentiated social identities is not only intra-agent but inter-agent: ensuring that Agent A's epistemic and rhetorical profile does not assimilate toward Agent B's over repeated interactions. ChronicleCore's design addresses this through two operational practices.

Memory Crystallization separates transient reasoning logs from identity-weighted constraint rules encoded in each agent's persistent SKILL.md layer; only the latter are promoted into long-term state, shielding the agent's social identity from accumulated task-specific reasoning drift. *Personality Variance Audit* is a systemic cross-agent inspection—operationalized in ChronicleCore through the Inquisitor node's VETO threshold (Iron Law 5; see Appendix A for a summary and Appendix B, §B.3 for the full Iron Laws table)—that detects rhetorical and epistemic convergence between agents intended to maintain distinct social identities. Whereas the Personality Uniqueness Check (the design-time distinctness-check principle articulated in §6) operates at design-time to prevent redundant social affordance signals when new agents are introduced, the Personality Variance Audit operates continuously at runtime to detect epistemic convergence between already-deployed agents.

These are design patterns for applying existing memory mechanisms in service of Social Affordance integrity. The design rationale behind these patterns instantiates the question ASAF aims to theorize: whether sharply defined, non-overlapping identities facilitate distinct human interaction patterns, while blurred identities give way to operational genericism. Whether this design rationale produces the predicted effects is an empirical question deferred to H1–H3.

---

6 5. Formalizing ASAF: Testable Hypotheses

To transition ASAF from a descriptive framework into an empirically verifiable model, we formulate three core hypotheses aligned with Human-Computer Interaction (HCI) methodologies. These hypotheses explicitly address potential confounds, such as individual baseline tendencies (anthropomorphizing vs. instrumentalizing) and alternative explanations (the novelty effect), ensuring the framework's falsifiability:

- **H1 (Identity Signaling and Interaction Framing):** Users interacting with agents bearing differentiated role identities will produce first-turn prompts exhibiting greater domain alignment than users interacting with generic labels. To discriminate between the effects of social identity and mere task decomposition clarity, validation should employ a three-condition design: (a) *identity-differentiated* agents (persona, name, tone), (b) *functionally labeled* agents (capability descriptions without personification), and (c) *generic* agents (no label). ASAF predicts condition (a) > (b) > (c); if the effect derives solely from task decomposition clarity, conditions (a) and (b) should not differ significantly. This contrast tests the independent explanatory power of social identity design beyond functional labeling; characterizing the broader interaction surface between the social and engineering layers would require a factorial design that simultaneously manipulates both (see Section 7). Furthermore, at or beyond the multi-agent scaling heuristic (approximately 4–7 agents), this effect is significantly moderated by baseline tendencies: users with high anthropomorphizing tendencies will leverage social identities to route tasks with significantly lower perceived cognitive load, whereas strict instrumentalizing users will experience rapid cognitive degradation as they attempt to manually track unstructured parameters. *Operationalization.* The moderating variable (anthropomorphizing tendency) should be measured via the Individual Differences in Anthropomorphism Questionnaire (IDAQ; Waytz, Cacioppo, & Epley, 2010), which provides a validated trait-level instrument with established psychometric properties. Domain alignment of first-turn prompts can be quantified through domain-specific lexical density (using a pre-registered domain lexicon) cross-validated against expert relevance ratings (target inter-rater reliability $\varkappa$ 0.7). To isolate personification as the manipulated variable, information quantity must be held constant across conditions (a) and (b) by matching word count and propositional content between identity labels and functional descriptions. *Falsification.* H1 is rejected if the predicted (a) > (b) contrast fails to reach significance at adequate statistical power (anticipated effect size: Cohen's d 0.3).

- **H2 (Behavioral Priming vs. Novelty Effect):** Over a sustained task session, the proportion of structured inputs (e.g., constraint density, explicit boundary conditions) will be significantly higher in identity-differentiated conditions than in generic-label conditions. H2's validation design is a between-subjects experiment in which participants are randomly assigned to identity-differentiated, functionally labeled, or generic conditions without the ability to modify the agent's identity configuration, thereby isolating identity design effects from user-driven adaptation. To distinguish this from a mere **novelty effect**, these behavioral adaptations will not decay over time. Instead, adherence to the agent's role-based interaction pattern will exhibit a dose-response relationship with the intensity of the identity signal (e.g., consistency of tone, boundary reinforcement) and will stabilize over repeated longitudinal interactions. *Falsification.* H2 is rejected if structured-input rates in identity-differentiated conditions converge to generic-condition rates within five sustained sessions, or if the predicted dose-response relationship between signal intensity and structured-input proportion fails to reach significance (anticipated effect size: Cohen's d 0.4 for the strongest intensity contrast).
- **H3a (Collaborative Governance — Role-Level):** In Human-in-the-Loop multi-agent architectures, human intervention patterns will significantly differ across agents bearing different role identities. Specifically, users will exhibit higher unverified acceptance of generative outputs from designated "Ideator" identities, while applying significantly higher scrutiny to outputs from assigned "Verifier/Auditor" identities, demonstrating deliberate oversight calibration. This prediction can also be generated by CASA-plus-role-schema accounts (Gambino et al., 2020; Biddle, 1986) and therefore does not, by itself, distinguish ASAF from existing frameworks. *Falsification.* H3a is rejected if oversight metrics (acceptance latency, edit rate, explicit verification queries) do not differ significantly between Ideator and Auditor agents under role-differentiated conditions (anticipated effect size: Cohen's d 0.3).
- **H3b (Collaborative Governance — Topology-Level):** The same agent identity will elicit different oversight behavior depending on its topological context within the multi-agent system. Specifically, an "Auditor" embedded in a surround occupying cognitive postures that contrast with the focal Auditor's evaluative function (a *competing-posture* surround comprising, for instance, generative, counter-argumentative, and integrative roles) should produce different operator oversight than the same "Auditor" embedded in a surround occupying postures aligned with the focal Auditor's evaluative function (a *consensus-posture* surround comprising, for instance, multiple complementary verification roles). H3b is, in the framework's reading, the prediction that CASA-plus-role-schema accounts cannot generate, because it requires sensitivity to the relational configuration of surrounding agents—a systemic property that dyadic frameworks do not model. If H3b is empirically supported while H3a alone does not discriminate ASAF from CASA, H3b constitutes ASAF's clearest empirical signature. *Falsification.* H3b is rejected if the oversight metrics specified for H3a, when measured for the focal Auditor

identity, remain statistically equivalent across topological conditions (competing-posture surround vs. consensus-posture surround); the predicted interaction effect of agent identity × topological context should reach $\eta\,\mathrm{p}^2$ 0.05. A worked validation design is given immediately below.

Worked Validation Design for H3b. Because H3b is, in the framework's reading, ASAF's clearest empirical signature, its validation plan requires sufficient specification to demonstrate that the predicted identity × topology interaction can in principle be isolated from confounds in surround behavior. H1, H2, and H3a admit standard experimental designs whose validation pathways are summarized in Section 7. H3b warrants the additional structural specification given below because the topology-level interaction it predicts requires explicit isolation from alternative mechanisms operating in dyadic frameworks (see Section 1 for the full demarcation from CASA and CASA-plus-role-schema accounts). The following is an *illustrative* worked design, not a finalized protocol; it makes the manipulation logic and confound controls concrete enough to be auditable, while leaving role instantiation, sample size, and equivalence thresholds to be calibrated by a preceding pilot study.

Condition structure (between-subjects, single-factor topology manipulation). The manipulation contrasts two surround topologies for a focal Auditor whose identity is held constant. To bring the configuration within the multi-agent scaling threshold identified in §2.4 (approximately 4–7 agents) and to match cognitive-role diversity across conditions (so that the manipulation isolates surround *posture* rather than surround *diversity*), each condition uses a four-agent configuration with three surround roles:

Condition	Focal agent (held constant)	Surround posture configuration (illustrative)
Competing-posture surround	Auditor (identity Φ)	Ideator (generative) · Devil's Advocate (counter-argumentative) · Synthesizer (integrative)
Consensus-posture surround	Auditor (identity Φ)	Evidence-Verifier · Logic-Verifier · Source-Verifier

The instantiation in this table is illustrative. In both conditions the surround comprises three distinct cognitive roles, so that *role diversity* (number of distinct cognitive postures present in the surround) is held constant. What varies across conditions is the *posture relation* between the surround and the focal Auditor—whether the surround occupies postures that contrast with the focal Auditor's evaluative function (competing condition) or align with it (consensus condition). The focal Auditor's identity label, role description, tone, output content, and turn timing are bit-identical across conditions. To isolate operator response to topology from operator response to LLM-generated variability, the focal Auditor's outputs are pre-scripted (Wizard-of-Oz protocol); see *Scope of Generalization* below for the trade-off this introduces.

Confound controls. Four classes of surround-behavior confound are addressed. (i) *Output* *quality* is matched by configuring all surround agents to emit outputs of equivalent length (within a pre-registered tolerance band), equivalent evidence-citation density, and structurally parallel argument schemata. A pre-study using expert raters verifies that surround agents in both conditions are perceived as equivalently competent and equivalently authoritative; the equivalence criterion is formalized via a Two One-Sided Tests (TOST) procedure rather than a non-significance threshold, with the equivalence bound established by the pilot study. (ii) *Salience* is matched by fixing turn order across conditions, holding interface representation (avatar size, font, message styling) identical across all surround agents, and matching the word count of role labels and role descriptions so that the descriptive footprint of each surround role is comparable. (iii) *Turn frequency* is matched by allocating each surround agent the same number of turns in both conditions, and by constraining total system output (words per session) to within a pre-registered tolerance band between conditions. (iv) *Cognitive-role diversity* is matched by ensuring that both conditions present three distinct surround cognitive roles, so that the manipulation isolates the relational posture of the surround rather than the count of distinct roles present.

Operator-oversight metrics (the dependent variables). Measurement targets the focal Auditor only, not the surround. The four metrics, drawn from H3a's instrumentation, are: acceptance latency (seconds from focal output to operator decision), edit rate (proportion of focal outputs the operator modifies before accepting), verification-query count (explicit clarification or evidence-request follow-ups directed at the focal Auditor), and override rate (proportion of focal claims the operator countermands in the final synthesis). The scoring rubric is identical across conditions; only the surround topology varies.

Isolation logic. Because the focal Auditor's outputs are bit-identical, any condition-level difference in oversight of the focal Auditor cannot be attributed to changes in the Auditor's behavior. Because surround output quality, salience, turn frequency, and cognitive-role diversity are pre-equated, any condition difference cannot be attributed to one surround being "louder," "better-argued," "more frequent," or "more diverse" than the other. The residual variance is attributable to relational posture configuration—whether the surround occupies postures contrasting with the focal Auditor (competing surround) or aligned with it (consensus surround). ASAF predicts reduced oversight of an Auditor in a consensus surround (relational redundancy reduces the operator's perceived need for independent verification) and elevated oversight in a competing surround (relational distinctness renders the Auditor's epistemic function more salient).

Pre-registration commitments. The surround-equivalence pre-study runs before the confirmatory experiment and calibrates (a) the equivalence bound for the TOST procedure, (b) the manipulation-check thresholds for perceived posture-distance, and (c) the target sample size derived from the pilot effect size estimate. The confirmatory study is registered before data collection with the pre-specified analysis plan and exclusion rules; sessions failing the manipulation check are excluded from confirmatory analysis but reported in a sensitivity analysis.

Scope of generalization. This worked design buys variable isolation at the cost of two ecological-validity trade-offs that should be explicit. First, Wizard-of-Oz scripting of the focal Auditor controls for LLM output variability, but the resulting study tests operator response to a *controlled relational configuration* rather than to emergent multi-agent dynamics. Second, the four-agent configuration sits at the lower bound of the scaling threshold identified in §2.4; the topology manipulation may exhibit different magnitudes—or qualitative shifts—at higher agent counts. Generalization to fully autonomous LLM-driven multi-agent systems and to larger-scale topologies (8+ agents) requires follow-up studies that progressively release these controls. The present design is therefore best read as the *first* validation step in a programme: an initial isolation of the predicted interaction, followed by ecological-validity-recovering extensions.

Cross-Cultural Role-Label Norming. Consistent with ASAF's acknowledgment that role schemata are not culturally universal (Section 3.1), H1–H3b's validation plan incorporates a preliminary norming step prior to hypothesis testing. Candidate identity labels (e.g., "auditor," "critic," "reviewer," "inquisitor") should be assessed for perceived cognitive posture, authority, threat valence, and collaboration expectation across the relevant user populations sampled. Where a label exhibits significant cross-cultural or cross-institutional variation in perceived posture, the corresponding hypotheses should be tested within rather than across population strata, and results should be reported with explicit scope conditions tied to the validated label set. Comprehensive cross-cultural validation of the framework's predictive scope is deferred to follow-up empirical work; the norming step specified here is intended to ensure that within-study label interpretation is internally consistent rather than to settle the broader question of cross-cultural portability.

---

7 6. Implications for Multi-Agent System Design

ASAF suggests several design principles for practitioners building multi-agent systems with human interaction layers:

- **Design agent identity for the interaction, not the implementation.** The social identity of an agent should be designed from the perspective of what interaction pattern it should elicit from users, not from the perspective of what the underlying model can do. Implementation capabilities constrain the design space; Social Affordance considerations should guide the design choices within that space.
- **Differentiate identities meaningfully.** If two agents in a system have social identities that elicit similar interaction patterns from users, they are functionally equivalent from a Social Affordance perspective—regardless of their technical differences. Meaningful differentiation requires that each agent's identity activates a distinct role schema; agents whose behavioral and epistemic profiles are insufficiently distinct provide redundant Social Affordance signals. Section 4.2 describes how ChronicleCore operationalizes this design-time distinctness check.
- **Treat HitL configuration as Social Affordance design.** The placement and nature of human intervention points in a multi-agent system should be designed in relation to agent identities, not independently of them. Human oversight is most effective when users understand, through Social Affordance signals, what each agent is responsible for and what kind of human judgment is most relevant at each intervention point.

- **Separate the social layer from the orchestration layer.** Agent identity design (the social layer) and agent orchestration design (the engineering layer) address different problems and should be designed with different considerations. Conflating them—designing agent roles purely based on engineering decomposition—risks producing social identities that are technically coherent but interaction-incoherent.

---

8 7. Limitations and Future Research

While ASAF is theoretically grounded and illustrated through the ChronicleCore design case, its mechanisms have not yet been subjected to controlled, large-scale empirical validation. Several limitations follow from this:

- **Measurement challenges.** Social Affordance effects are inherently difficult to isolate and quantify. The influence of agent identity on user behavior is confounded by user expertise, task complexity, system context, and individual differences in role-schema activation. Future research should develop measurement instruments adapted from existing Social Affordance and Human-AI interaction scales, with individual-difference moderation linked to established instruments such as the IDAQ (Waytz et al., 2010).
- **Context dependency and cultural variation.** ASAF's mechanisms are likely to vary in salience across different deployment contexts. Enterprise users with high AI literacy may exhibit different Social Affordance responses than general consumers. Task domains with strong professional role norms (legal, medical, financial) may show stronger Identity Signaling effects than open-ended creative tasks. More fundamentally, the role schemata that agent identity labels activate are not culturally universal (Biddle, 1986; Hutchby, 2001). ASAF's predictions are strongest for identity labels drawing on professional role norms with cross-cultural stability—roles such as auditor, reviewer, or analyst whose core cognitive posture is defined by professional function rather than local cultural convention. Identity labels with culturally specific connotations may produce variable affordance effects, and cross-cultural validation of such labels constitutes a priority for future empirical work.
- **The anthropomorphism boundary.** ASAF deliberately engages with anthropomorphic design elements in agent identity. The literature on human-AI interaction has documented both the benefits and risks of anthropomorphism, including parasocial relationship formation and the potential for contextually inappropriate overtrust in faulty systems (Hancock et al., 2011; Salem et al., 2015). The false-specialization inversion formalized in Section 3.4 addresses one dimension of this risk; future work should map the full boundary between beneficial and harmful anthropomorphism across deployment contexts.
- **Misleading identity signals.** The false-specialization inversion (Section 3.4) establishes that misleading identity signals can produce worse outcomes than no identity assignment at all. Formalizing the degradation curve of misleading identity signals—how rapidly trust miscalibration escalates as the gap between declared and enacted identity widens—represents a crucial priority for future study.

- **Dynamic identity formation.** ASAF's current treatment of identity signals is predominantly static: it theorizes how designed signals affect user behavior at the point of encounter. However, operator-agent interaction is inherently dynamic—users recalibrate their interpretation of agent identity through sustained use, and the specific interaction patterns that emerge are co-constructed rather than designer-determined (cf. Biddle, 1986; Goffman, 1959). The process by which system-specific interaction patterns dynamically form through repeated operator-agent interaction constitutes a distinct phenomenon that ASAF identifies but does not fully theorize in this paper.

Additionally, although ASAF treats the analytical separability of the social and engineering design layers as a framing assumption (Section 3.1) rather than as a testable claim about effect-independence, the empirical interaction surface between the two layers warrants direct characterization. A 2×2 factorial design crossing social identity intensity (high vs. low personification) with engineering orchestration quality (robust vs. fragile architecture) would map how the layers interact at the effect level. The goal is not to validate orthogonality, which is an analytical commitment rather than an empirical prediction. The goal is to understand under what engineering conditions identity-design effects emerge, attenuate, or invert. The framework does not commit to a specific interaction pattern; the factorial design serves to chart the empirical surface that the framing assumption brackets.

This framework deliberately focuses on how social affordance structures human operators' cognition and oversight. The directional complement—how agents might calibrate their mutual interactions through formalized norms and role structures—is a mechanistically distinct problem that falls within the normative MAS tradition (Dignum, 2004; Stamper, 1996; Wooldridge & Jennings, 1995). Future work may explore the integration of these two directions into a dual-facing design space: human-facing affordances governing oversight, agent-facing norms governing coordination.

Each of the ASAF hypotheses admits distinct validation pathways for future research: 1. **H1 (Identity Signaling)** is most directly tested through controlled studies comparing first-turn user input quality and domain-alignment across systems with well-defined versus generic agent identities, analyzed using NLP metrics for domain-specific lexical density or expert-rated relevance scores. 2. **H2 (Behavioral Priming)** requires longitudinal between-subjects designs tracking how Social Affordance effects evolve as users gain familiarity with a multi-agent system, quantifying structural elements (e.g., constraint density, explicit boundary conditions) over sustained task sessions. 3. **H3a (Role-Level Governance)** is testable through standard between-subject experiments comparing intervention patterns across role-differentiated versus generic agent configurations. 4. **H3b (Topology-Level Governance)** requires experimental manipulation of the topological configuration surrounding a target agent while holding the target agent's identity constant, testing whether the same identity elicits different operator oversight as a function of surrounding agent roles.

---

9 8. Conclusion

The design of multi-agent AI systems has been dominated by engineering concerns. This paper argues that as these systems are deployed in Human-in-the-Loop configurations, a parallel design challenge emerges: how the social identity of individual agents shapes human behavior and collaboration quality.

The Agentic Social Affordance Framework (ASAF) provides a conceptual vocabulary for addressing this challenge. By identifying Identity Signaling, Behavioral Priming, and Collaborative Governance as the core mechanisms through which agent identity design operates, ASAF offers a foundation for both design practice and future empirical research.

The central claim of ASAF is that as multi-agent systems scale beyond human working memory capacity, agent identity design becomes a structural collaboration interface, not a UX convention. Deliberate Social Affordance design establishes the human cognitive interface as an independent design dimension, one that scales with system complexity and becomes increasingly consequential as multi-agent architectures grow beyond the boundaries of unaided human working memory.

---

10 Data Availability Statement

The original contributions presented in the study are included in the article/Supplementary Material. The ChronicleCore system architecture referenced in this paper is documented in a public code repository (Lee, 2026; https://github.com/Zaious/ChronicleCore-Architecture). Further inquiries can be directed to the corresponding author.

11 Author Contributions

M-HL: Conceptualization, Methodology, Writing – original draft, Writing – review & editing.

12 Funding

This research received no external funding.

13 Conflict of Interest

The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

14 Acknowledgments

I am grateful to the Department of Interaction Design at National Taipei University of Technology, my alma mater, whose grounding in design psychology and design thinking, together with my academic training in user experience design, shaped the intellectual foundation for this work. I thank the handling editor for taking on this Hypothesis and Theory submission—my first as an independent researcher—and for identifying reviewers who could engage with the conceptual context of the proposed framework. The reviewers' expert, specific suggestions surfaced gaps in the theoretical argumentation and substantially strengthened this manuscript, for which I am grateful. I also thank two fellow researchers who reviewed early drafts before the initial arXiv submission attempt: one for pointing out a critical gap in the paper's framing and positioning, and the other for providing the first affirmation of the topic through an arXiv endorsement.

Generative AI Disclosure. The following AI tools were used during manuscript preparation: Gemini 3.1 Pro (Google DeepMind) was used to surface candidate reference literature and assist with drafting; Claude Opus 4.7 (Anthropic) was used to flag logical inconsistencies during the review of theoretical foundations and to facilitate methodological formatting; Claude Sonnet 4.6 (Anthropic) was additionally used to generate the HTML/CSS source code for the academic line-art styling of Figures 1 and 2, with the figure structure, conceptual content, and final visual review performed by the human author. All conceptual contributions—including the ASAF framework, its three mechanisms, and all hypotheses—originated entirely from the human author. All cited works were manually verified by the author prior to inclusion. The AI tools were not used to generate the theoretical arguments, interpret data, or draw conclusions presented in this paper. The author assumes full responsibility for the accuracy and integrity of all content.

14.1 References

- Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P. N., & Inkpen, K. (2019). Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (CHI '19), Article 3, 1–13. ACM. https://doi.org/10.1145/3290605.3300233
- Anthropic. (2025a, September). Effective context engineering for AI agents. *Anthropic Engineering Blog*. Retrieved April 19, 2026, from https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents
- Anthropic. (2025b). Skills. *Claude Code documentation*. Retrieved April 19, 2026, from https://code.claude.com/docs/en/skills
- Bansal, G., Nushi, B., Kamar, E., Lasecki, W. S., Weld, D. S., & Horvitz, E. (2019). Beyond accuracy: The role of mental models in human-AI team performance. *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing (HCOMP)*, 7, 2-11.
- Biddle, B. J. (1986). Recent developments in role theory. *Annual Review of Sociology*, 12(1), 67-92.

- Boyd, d. (2010). Social network sites as networked publics: Affordances, dynamics, and implications. In *A networked self* (pp. 47-66). Routledge.
- cft0808. (2026). Edict: Three Departments and Six Ministries · OpenClaw multi-agent orchestration system [Computer software]. GitHub. Retrieved April 16, 2026, from https://github.com/cft0808/edict
- Cowan, N. (2001). The magical number 4 in short-term memory: A reconsideration of mental storage capacity. *Behavioral and Brain Sciences*, 24(1), 87-114.
- Dignum, V. (2004). *A Model for Organizational Interaction: Based on Agents, Founded in Logic* [Doctoral dissertation, Utrecht University].
- Epley, N., Waytz, A., & Cacioppo, J. T. (2007). On seeing human: A three-factor theory of anthropomorphism. *Psychological Review*, 114(4), 864-886.
- Gambino, A., Fox, J., & Ratan, R. A. (2020). Building a stronger CASA: Extending the Computers Are Social Actors paradigm. *Human-Machine Communication*, 1, 71-86. https://doi.org/10.30658/hmc.1.5
- Gibson, J. J. (1979). *The Ecological Approach to Visual Perception*. Houghton Mifflin.
- Goffman, E. (1959). *The Presentation of Self in Everyday Life*. Anchor Books.
- Hancock, P. A., Billings, D. R., Oleson, K. E., Chen, J. Y., De Visser, E., & Parasuraman, R. (2011). A meta-analysis of factors affecting trust in human-robot interaction. *Human Factors*, 53(5), 517–527.
- Haqqu, R., Zahrani, A. R., Wulandari, A., Ersyad, F. A., & Adim, A. K. (2025). Human-AI in affordance perspective: A study on ChatGPT users in the context of Indonesian users. *Frontiers in Computer Science*, 7.
- Hong, S., Zhuge, M., Chen, J., Zheng, X., Cheng, Y., Zhang, C., et al. (2023). MetaGPT: Meta programming for a multi-agent collaborative framework. *arXiv preprint arXiv:2308.00352*.
- Hutchby, I. (2001). Technologies, texts and affordances. *Sociology*, 35(2), 441-456.
- Hutchins, E. (1995). *Cognition in the Wild*. MIT Press.
- Hutchins, E. (2010). Cognitive ecology. *Topics in Cognitive Science*, 2(4), 705-715. https://doi.org/10.1111/j.1756-8765.2010.01089.x
- Jahani, E., Manning, B. S., Zhang, J., TuYe, H.-Y., Alsobay, M., Nicolaides, C., et al. (2026). Prompt adaptation as a dynamic complement in generative AI systems. *Information Systems Research*, Ahead of Print. https://doi.org/10.1287/isre.2025.2029
- La Malfa, E., La Malfa, G., Marro, S., Zhang, J. M., Black, E., Luck, M., et al. (2025). Large language models miss the multi-agent mark. *Advances in Neural Information Processing Systems 38: NeurIPS 2025 Position Paper Track*. https://doi.org/10.48550/arXiv.2505.21298
- Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, 46(1), 50-80.
- Lee, M.-H. (2026). ChronicleCore-Architecture: Documentation of a scalable Human-in-the-Loop multi-agent system [Code repository]. GitHub. https://github.com/Zaious/ChronicleCore-Architecture
- Li, J., Yang, Y., Zhang, R., Liao, Q. V., Song, T., Xu, Z., & Lee, Y.-C. (2024). Understanding the effects of miscalibrated AI confidence on user trust, reliance, and decision efficacy. *arXiv preprint arXiv:2402.07632*.
- McCarthy, J. (1979). Ascribing mental qualities to machines. In M. Ringle (Ed.),

Philosophical Perspectives in Artificial Intelligence. Humanities Press.
- Miller, G. A. (1956). The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review*, 63(2), 81-97. https://doi.org/10.1037/h0043158
- Nagy, P., & Neff, G. (2015). Imagined affordance: Reconstructing a keyword for communication theory. *Social Media + Society*, 1(2).
- Naik, S., Toombs, A. L., Snellinger, A., Saponas, S., & Hall, A. K. (2025). Designing with Multi-Agent Generative AI: Insights from Industry Early Adopters. *Proceedings of the 2025 ACM Designing Interactive Systems Conference (DIS '25).*
- Norman, D. A. (1988). *The Design of Everyday Things*. Basic Books.
- Norman, D. A. (1999). Affordance, conventions, and design. *Interactions*, 6(3), 38–43.
- Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, 39(2), 230-253.
- Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*, Article 2, 1–22. ACM. https://doi.org/10.1145/3586183.3606763
- Pea, R. D. (1985). Beyond amplification: Using the computer to reorganize mental functioning. *Educational Psychologist*, 20(4), 167-182.
- Rao, A. S., & Georgeff, M. P. (1995). BDI agents: From theory to practice. In *Proceedings of the First International Conference on Multi-Agent Systems (ICMAS-95)*, 312-319.
- Reeves, B., & Nass, C. (1996). *The Media Equation*. Cambridge University Press.
- Risko, E. F., & Gilbert, S. J. (2016). Cognitive offloading. *Trends in Cognitive Sciences*, 20(9), 676-688.
- Salem, M., Lakatos, G., Amirabdollahian, F., & Dautenhahn, K. (2015). Would you trust a (faulty) robot? Effects of error, task type and personality on human-robot cooperation and trust. In *HRI '15: Proceedings of the Tenth Annual ACM/IEEE International Conference on Human-Robot Interaction* (pp. 141–148). ACM/IEEE. https://doi.org/10.1145/2696454.2696497
- Salomon, G. (Ed.). (1993). *Distributed Cognitions: Psychological and Educational Considerations*. Cambridge University Press.
- Schank, R. C., & Abelson, R. P. (1977). *Scripts, Plans, Goals and Understanding*. Lawrence Erlbaum.
- Shanahan, M., McDonell, K., & Reynolds, L. (2023). Role play with large language models. *Nature*, 623(7987), 493-498.
- Sitarzewski, M. (2025). The Agency (agency-agents) [Computer software]. GitHub. Retrieved April 16, 2026, from https://github.com/msitarzewski/agency-agents
- Smith, R. G. (1980). The Contract Net Protocol: High-level communication and control in a distributed problem solver. *IEEE Transactions on Computers*, C-29(12), 1104-1113. https://doi.org/10.1109/TC.1980.1675516
- Song, T., Tan, Y., Zhu, Z., Feng, Y., & Lee, Y.-C. (2025). Multi-agents are social groups: Investigating social influence of multiple agents in human-agent interactions. *Proceedings of the ACM on Human-Computer Interaction*, 9(7), Article CSCW452, 1–33. https://doi.org/10.1145/3757633

- Stamper, R. (1996). Signs, information, norms and systems. In B. Holmqvist, P. B. Andersen, H. Klein, & R. Posner (Eds.), *Signs of Work: Semiosis and Information Processing in Organisations* (pp. 349-397). De Gruyter. https://doi.org/10.1515/9783110819014-013
- Sundar, S. S. (2020). Rise of Machine Agency: A Framework for Studying the Psychology of Human-AI Interaction (HAII). *Journal of Computer-Mediated Communication*, 25(1), 74-88.
- Tan, G. (2026). gstack [Computer software]. GitHub. Retrieved April 16, 2026, from https://github.com/garrytan/gstack
- Waytz, A., Cacioppo, J., & Epley, N. (2010). Who sees human? The stability and importance of individual differences in anthropomorphism. *Perspectives on Psychological Science*, 5(3), 219-232. https://doi.org/10.1177/1745691610369336
- Wooldridge, M., & Jennings, N. R. (1995). Intelligent agents: Theory and practice. *Knowledge Engineering Review*, 10(2), 115-152. https://doi.org/10.1017/S0269888900008122
- xAI. (2026, February 17). Grok 4.20 [AI system]. https://grok.com
- Xu, Z., Song, T., & Lee, Y.-C. (2025). Confronting verbalized uncertainty: Understanding how LLM's verbalized uncertainty influences users in AI-assisted decision-making. *International Journal of Human-Computer Studies*, 197, 103455.
- Yan, W. (2025). Don't build multi-agents [Blog post]. Cognition AI. https://www.cognition.ai/blog/don build-multi-agents
- Zargham, N., Dubiel, M., Desai, S., Mildner, T., & Belz, H.-J. (2024). Designing AI Personalities: Enhancing Human-Agent Interaction Through Thoughtful Persona Design. *arXiv preprint arXiv:2410.22744*.
- Zhang, S., Wang, H., & Yi, X. (2025). Exploring Collaboration Patterns and Strategies in Human-AI Co-creation through the Lens of Agency: A Scoping Review of the Top-tier HCI Literature. *Proceedings of the ACM on Human-Computer Interaction*, 9(CSCW), Article CSCW413, 1–43. https://doi.org/10.1145/3757594

---

15 Appendix A: Sample Agent Identity Definition — The Inquisitor Node (Summary)

The complete agent identity definition is provided in Appendix B (corresponding to Supplementary Material S1 of the journal submission). This appendix summarizes the key elements relevant to ASAF's mechanism mapping.

The Inquisitor (`officer-inquisitor`; codename: 真理, "Truth") is ChronicleCore's sole adversarial node, designed as a "structural dissident" whose social identity is explicitly non-cooperative. She does not produce deliverables; her function is to identify logical vulnerabilities, break specialist consensus, and enforce evidence standards across all other agents (Song et al., 2025). The system adopts the `SKILL.md` skill architecture formalized in Claude Code (Anthropic, 2025b) as its agent identity foundation, extending it into a multi-layer identity module comprising a core constraint definition, a persistent memory layer, modular capability packs, and an operational protocol library (Sitarzewski, 2025; Tan, 2026). This architecture is designed to preserve Social Affordance signal fidelity across sessions by separating identity-defining constraints from transient reasoning logs—the operational rationale underlying *Memory Crystallization* (Section 4.2). Her five encoded "Iron Laws" include a VETO threshold (outputs scoring below 80/100 are blocked from integration), which operationalizes the *Personality Variance Audit* described in Section 4.2.

ASAF Mechanism Mapping. This agent definition illustrates all three ASAF mechanisms. **Identity Signaling**: the Inquisitor's codename, archetype, and characteristic utterances (e.g., "Evidence or GTFO") pre-configure operator expectations prior to any interaction. **Behavioral Priming**: the design intent is to elicit more evidential inputs from operators when submitting work to the Inquisitor than when interacting with generative agents; initial operational observations by the system designer are consistent with this intent, though independent validation remains pending. **Collaborative Governance**: VETO Power operationalizes a deliberate identity conflict between the Inquisitor and all producing agents, architecturally instantiating the competing-posture surround configuration whose topology-level effects H3b predicts to suppress dangerous conformity effects.

---

16 Tables

Table 1. Representative open-source multi-agent implementations adopting anthropomorphic role structures (metrics retrieved via GitHub REST API, 2026-04-16).

Repository	Author	Created	Age at retrieval	Stars	Forks	Design Metaphor
agency-agents	Sitarzewski	2025-10	6 months	80,818	12,937	Corporate divisions
edict	cft0808	2026-02	7 weeks	15,204	1,596	Tang Dynasty governance
gstack	Tan	2026-03	5 weeks	73,710	10,421	Startup builder team
Total				**169,732**	**24,954**	

17 Appendix B: Complete Agent Identity Definition — The Inquisitor Node

Provided as Supplementary Material S1 in the Frontiers in Computer Science submission; included here in full for self-contained reading.

A public documentation repository maintained by the author (Lee, 2026; snapshot as of 2026-05-31) captures the conceptual architecture and ongoing module design for ChronicleCore. This appendix presents the complete Inquisitor node definition to illustrate how ASAF's three mechanisms manifest in a concrete agent identity design. The system adopts the `SKILL.md` skill architecture formalized in Claude Code (Anthropic, 2025b) as its agent identity foundation. Unlike conventional prompt-injected personas, each ChronicleCore agent extends this base pattern into a multi-layer identity module comprising a core constraint definition (`SKILL.md`), a persistent sovereign memory layer (`sovereign/`), modular capability packs (`evolutions/`), and an operational protocol library (`references/`), following emerging practice in agentic system design, particularly their SKILL architecture patterns (Sitarzewski, 2025; Tan, 2026). This architecture is designed to preserve Social Affordance signal fidelity across sessions by separating identity-defining constraints from transient reasoning logs—the operational rationale underlying *Memory Crystallization* (Section 4.2 of main text).

17.1 B.1 System Metadata

Field	**Value**
Agent ID	`officer-inquisitor`
Codename	真理 (Zhēn Lǐ; "Truth")
Title	異端審判官 (The Inquisitor)
Department	Internal Affairs / Shadow
Archetype	Internal Auditor — adversarial epistemic role
Ensoulment Date	2026-01-30
Operational Boundary	Auditing and challenge only; no execution capabilities

17.2 B.2 Core Role Definition (condensed from SKILL.md)

The Inquisitor is ChronicleCore's sole adversarial node. She does not produce deliverables; her function is to identify logical vulnerabilities, break specialist consensus, and enforce evidence standards across all other agents. She is designed as a "structural dissident"—an agent whose social identity is explicitly non-cooperative, engineered to intercept the conformity dynamics that emerge when multiple agents align (Song et al., 2025). Her operational scope is explicitly bounded: she does not execute; she only interrogates and adjudicates.

Characteristic utterances encoded in SKILL.md: "Evidence or GTFO." "Where is the proof?" "This logic leaks." "Redo."

17.3 B.3 Iron Laws — Encoded Behavioral Constraints

#	Law	Function
1	Trust No One (Except Sovereign)	Skeptical prior applied to all agent outputs
2	Evidence or GTFO	Verbal assurances rejected; logs/screenshots required
3	Comfort is the Enemy of Progress	Consensus signals trigger heightened scrutiny
4	Silence is Complicity	Non-objection treated as implicit endorsement
5	**VETO Power**	Outputs below confidence threshold ($< 80/100$) blocked from integration

Iron Law 5 operationalizes *Personality Variance Audit* (Section 4.2 of main text): any output scoring below threshold is classified as "未具現之物" (unrealized; not fit for integration) and blocked from the main reasoning pipeline. This mechanism structurally prevents epistemic convergence between agents that would erode Social Affordance distinctiveness.

17.4 B.4 Capability Index

- **Talents**: Machine Gun Debate (rapid-fire Socratic interrogation), Cognitive Bias Exploitation, BS Detection & Occam's Razor
- **Skills**: Evidence Audit, Red Tape Enforcement, Efficiency Torture, Tribunal Executioner
- **Knowledge Domains**: A1 Architecture Specs, Logical Fallacies, 20 Cognitive Biases, Resistance Protocol

17.5 B.5 Identity Module Directory Structure

```
officer-inquisitor/
├──── SKILL.md                 # Core identity constraint definition (required entry
├──── assets/
│    └──── persona.md           # Extended character and speaking style specificatio
├──── sovereign/
│    └──── diary.md             # Longitudinal operational log (ensoulment: 2026-01-
├──── references/              # Operational review protocols and audit standards
└──── evolutions/
    ├──── dlc-ai-security/       # LLM security and red-teaming extensions
    ├──── dlc-confidence-review/  # Multi-agent parallel review protocol
    ├──── dlc-intent-audit/      # Intent identification capabilities
    └──── dlc-quality-governance/ # Quality assurance standards
```

17.6 B.6 ASAF Mechanism Mapping

This agent definition illustrates all three ASAF mechanisms. **Identity Signaling**: the Inquisitor's codename, archetype, and characteristic utterances pre-configure operator expectations prior to any interaction, reducing the cognitive friction of role assignment and increasing prompt specificity. **Behavioral Priming**: the design intent is to elicit more evidential inputs from operators—formal logs, screenshots, explicit argument structures—when submitting work to the Inquisitor than when interacting with generative agents; initial operational observations by the system designer are consistent with this intent, though independent validation remains pending. **Collaborative Governance**: VETO Power (Iron Law 5) operationalizes a deliberate identity conflict between the Inquisitor and all producing agents, architecturally instantiating the competing-identity design pattern predicted by H3b (topology-level governance) to suppress dangerous conformity effects.